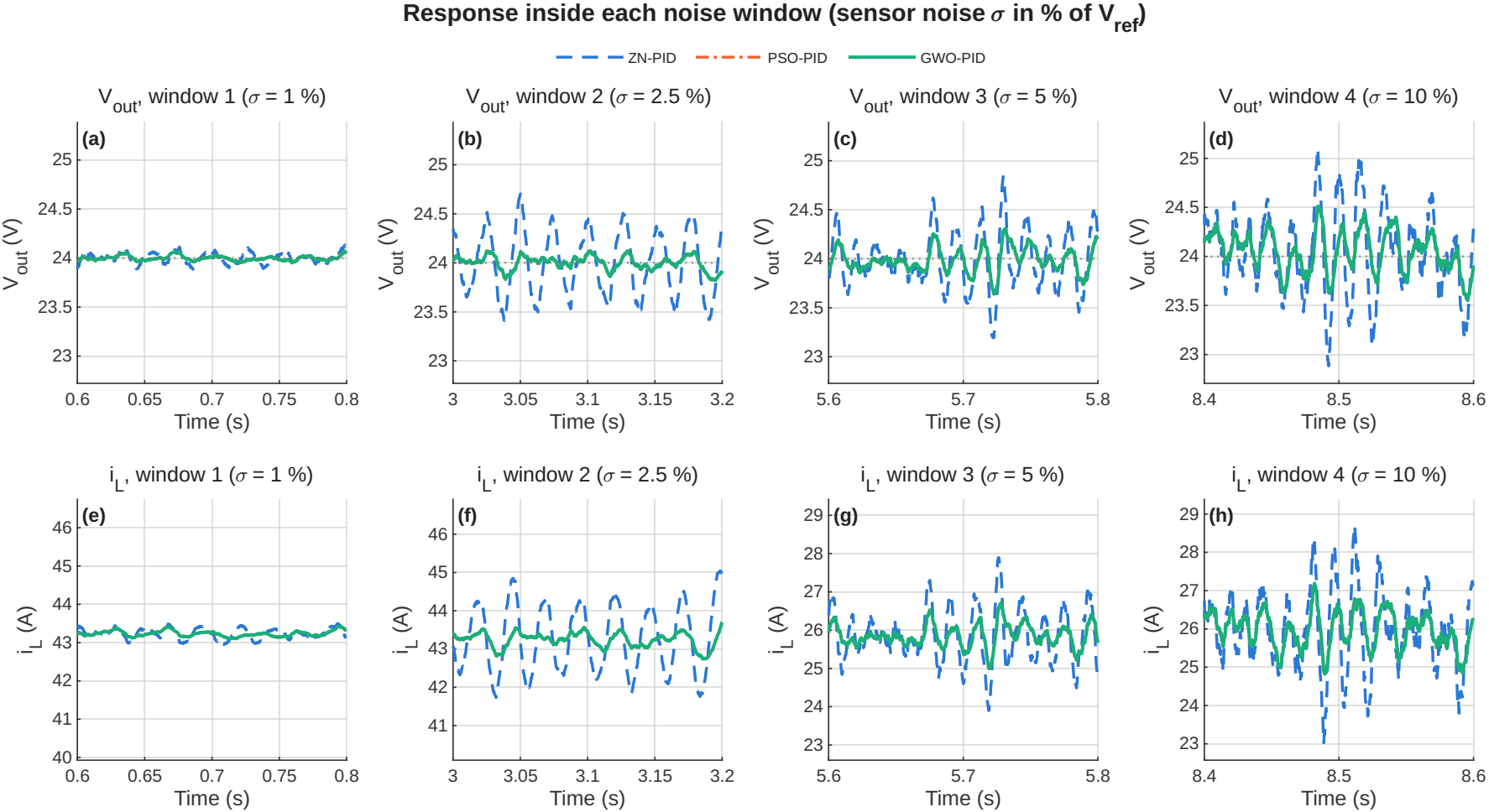

Response inside each noise window (sensor noise σ in % of V_ref)
ZN-PID
PSO-PID
GWO-PID
V_out, window 1 (σ = 1 %)
V_out, window 2 (σ = 2.5 %)
V_out, window 3 (σ = 5 %)
V_out, window 4 (σ = 10 %)
i_L, window 1 (σ = 1 %)
i_L, window 2 (σ = 2.5 %)
i_L, window 3 (σ = 5 %)
i_L, window 4 (σ = 10 %)
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
V_out (V)
i_L (A)
Time (s)